

Session 13: File HandlingTopics:Open, write, read, close files.Modes: r, w, a.Example: Write and read “daily goals.txt.”

Tasks

- 1) Create a file called playlist.txt and write the names of your top 3 favorite songs from Spotify into it using write mode (w).**

Code-

```
#include <stdio.h>

int main() {
    FILE *fptr = fopen("playlist.txt", "w");

    if (fptr == NULL) {
        printf("Error opening file!\n");
        return 1;
    }

    fprintf(fptr, "Blinding Lights\n");
    fprintf(fptr, "Perfect\n");
    fprintf(fptr, "Shape of You\n");

    fclose(fptr);

    printf("Songs written to playlist.txt successfully!\n");

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc file1.c -o file1.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\file1.exe
Songs written to playlist.txt successfully!
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

2) Open playlist.txt in read mode (r) and display each song name on a separate line in the console.

Code-

```
#include <stdio.h>
```

```
int main() {
```

```
    char song[100];
```

```
    FILE *fptr = fopen("playlist.txt", "r");
```

```
    if (fptr == NULL) {
```

```
        printf("Error opening file! Make sure playlist.txt exists.\n");
```

```
        return 1;
```

```
    }
```

```
    printf("Your Spotify Playlist:\n");
```

```
    while (fgets(song, sizeof(song), fptr) != NULL) {
```

```
        printf("%s", song);
```

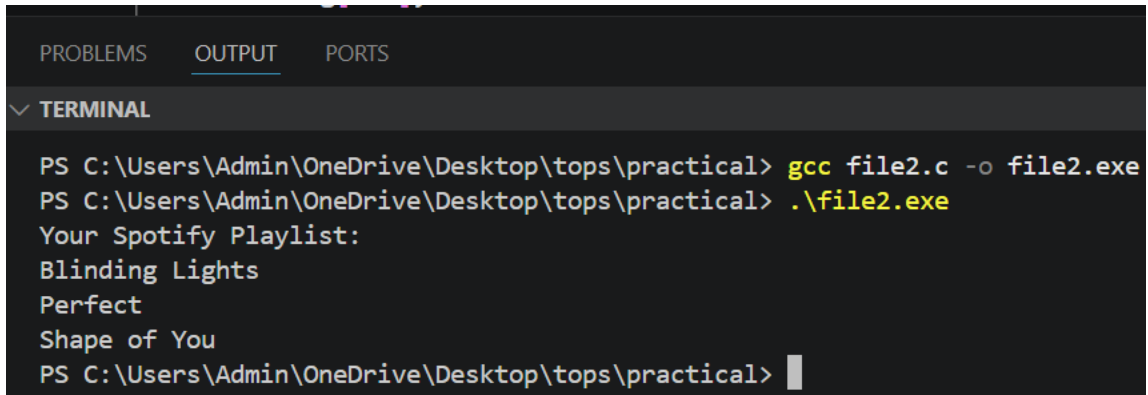
```
    }
```

```
    fclose(fptr);
```

```
    return 0;
```

```
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc file2.c -o file2.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\file2.exe
Your Spotify Playlist:
Blinding Lights
Perfect
Shape of You
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

The image shows a screenshot of a terminal window with a dark background. At the top, there are three tabs: 'PROBLEMS', 'OUTPUT', and 'PORTS'. The 'OUTPUT' tab is selected and underlined. Below the tabs, there is a section header '✓ TERMINAL'. The terminal content shows a PowerShell prompt 'PS C:\Users\Admin\OneDrive\Desktop\tops\practical>' followed by the command 'gcc file2.c -o file2.exe'. The next line shows the prompt again followed by the command '.\file2.exe'. The output of the program is displayed as 'Your Spotify Playlist:' followed by three lines of song titles: 'Blinding Lights', 'Perfect', and 'Shape of You'. The final line shows the prompt again with a cursor, indicating the program has finished execution.

3) Add two more song names to playlist.txt without deleting the existing ones by opening the file in append mode (a).

Code-

```
#include <stdio.h>
```

```
int main() {
```

```
    FILE *fptr = fopen("playlist.txt", "a");
```

```
    if (fptr == NULL) {
```

```
        printf("Error opening file!\n");
```

```
        return 1;
```

```
    }
```

```
    fprintf(fptr, "Levitating\n");
```

```
    fprintf(fptr, "Bad Habits\n");
```

```
    fclose(fptr);
```

```
    printf("Two new songs added to playlist.txt successfully!\n");
```

```
    return 0;
```

```
}
```

Output:

```
PROBLEMS  OUTPUT  PORTS
▼ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc file3.c -o file3.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\file3.exe
Two new songs added to playlist.txt successfully!
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

- 4) Write a program that reads all song names from playlist.txt and prints only those that contain the word 'love' (case-insensitive).

Hint: Use the 'in' keyword or equivalent string method for filtering.

Code-

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

int containsIgnoreCase(const char *text, const char *word) {

    char textCopy[100], wordCopy[100];

    int i;

    for (i = 0; text[i]; i++)

        textCopy[i] = tolower((unsigned char)text[i]);

    textCopy[i] = '\0';

    for (i = 0; word[i]; i++)

        wordCopy[i] = tolower((unsigned char)word[i]);

    wordCopy[i] = '\0';

    return strstr(textCopy, wordCopy) != NULL;

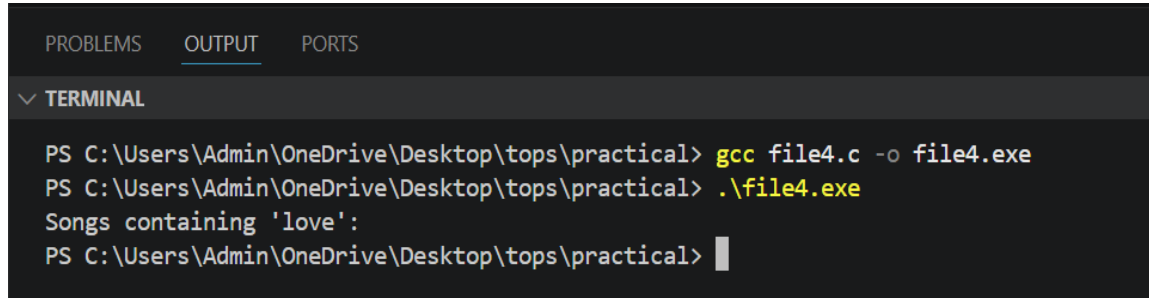
}
```

```
int main() {  
    char song[100];  
  
    FILE *fptr = fopen("playlist.txt", "r");  
  
    if (fptr == NULL) {  
        printf("Error opening file! Make sure playlist.txt exists.\n");  
        return 1;  
    }  
  
    printf("Songs containing 'love':\n");  
  
    while (fgets(song, sizeof(song), fptr) != NULL) {  
  
        song[strcspn(song, "\n")] = '\0';  
  
        if (containsIgnoreCase(song, "love")) {  
            printf("%s\n", song);  
        }  
    }  
  
    fclose(fptr);  
}
```



```
    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc file4.c -o file4.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\file4.exe
Songs containing 'love':
PS C:\Users\Admin\OneDrive\Desktop\tops\practical>
```

